Maharashtra Education Society's
Abasaheb Garware College (Autonomous), Pune
M.Sc. (Computer Science) Part II Semester III
SUBJECT: CS-611-MJP- Lab Course Based on Big Data

Time: 3 hours Max. Marks: 25

1. Write Hadoop shell commands to do the following [2]
   a. Create a new directory called data in HDFS.
   b. Upload a file called sample.txt from the local file system to the data directory in HDFS.
   c. List the files in the data directory.
   d. Move the file sample.txt to another directory in HDFS.

2. Consider the Movie Data set. Write a Map Reduce program to find average rating per user. [10]

3. Solve the following using HIVE [10]

   Write HQL queries for the following
   a. Create a database with the name "Udetails"
   b. Use the above created Database.
   c. Create a table UserData with the fields (uid, name, age, gender, occupation, zipcode) where fields are terminated by " , ".
   d. Load data from local storage in the path
   e. /home/abc/data/UData.txt' in the above table
   f. Find number of users for each age
   g. Add a column email_id to the above table.

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. Create a new directory called data in HDFS.
   b. Upload a file called sample.txt from the local file system to the data directory in HDFS
   c. Copy the sample.txt file to a new file called sample_copy.txt in the same directory.
   d. List the contents of the directory in HDFS.

2. Consider the Movie Data set. Write a Map Reduce program to find number of ratings per rating value. [10]

3. Solve the following using HIVE [10]

   Write HQL queries for the following
   1. Create a database with the name "Udetails"
   2. Use the above created Database.
   3. Create a table UserData with the fields (uid, name, age, gender, occupation, zipcode) where fields are terminated by " , ".
   4. Load data from local storage in the path /home/abc/data/UData.txt' in the above table
   5. Find average age of all females.
   6. Find number of users who are "IT Professionals"

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. Create a new directory called data in HDFS.
   b. Upload a file called sample.txt from the local file system to the data directory in HDFS
   c. Delete the sample_copy.txt file.
   d. List the directory contents in HDFS

2. Consider the Movie Data set. Write a Map Reduce program to find number of movies watched by each user. [10]

3. Solve the following using HIVE [10]
   Write HQL queries for the following
   1. Create a database with the name "Udetails"
   2. Use the above created Database.
   3. Create a table UserData with the fields (uid, name, age, gender, occupation, zipcode) where fields are terminated by " , ".
   4. -Load data from local storage in the path /home/abc/data/UData.txt' in the above table
   5. Find average age of all females.
   6. Find number of users for each age

4. Viva [5]

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1. Write Hadoop shell commands to do the following [2]
   a. Create a new directory called "test".
   b. Change into the "test" directory.
   c. Create a subdirectory called "subtest".
   d. List the files in the "subtest" directory
   .
2. Consider the Friends Data set. Write a Map Reduce program to find number of friends per user. [10]

3. Solve the following using HIVE-- [10]
   Write HQL queries for the following
   1. Create a database with the name "Udetails"
   2. Use the above created Database.
   3. Create a table UserData with the fields (uid, name, age, gender, occupation, zipcode) where fields are terminated by ",".
   4. Load data from local storage in the path /home/abc/data/UData.txt' in the above table
   5. Create the same table for UserData partitioned on gender
   6. Find number of users for each age

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. List the files in the /user directory.
   b. Create a new directory called navigation.
   c. Change into the navigation directory.
   d. Create a subdirectory called subnav.

2. Consider the Friends Data set. Write a Map Reduce program to find average number friends for each age value. [10]

3. Solve the following using HIVE [10]
   Write HQL queries for the following
   1. Create a database with the name IRating
   2. Use the above created database
   3. Create table Rating with the fields(userid, item_id, rating, timestamp) where fields are terminated by '\t'
   4. Load data from local storage in the path home/user/data/Item_Rating"
   5. Display first 20 rows of the table
   6. Find the total number of ratings for each user.

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. Create a new file called file1.txt in HDFS.
   b. Copy file1.txt to a new file called file2.txt.
   c. Copy file2.txt to a new directory called copydir.
   d. Verify the file contents in copydir.

2. Consider the Friends Data set. Write a Map Reduce program to find agewise number of friends sorted on age. [10]

3. Solve the following using HIVE [10]
   Write HQL queries for the following
   1. Create a database with the name IRating
   2. Use the above created database
   3. Create table Rating with the fields(userid, item_id, rating, timestamp) where fields are terminated by '\t'
   4. Load data from local storage in the path home/user/data/Item_Rating"
   5. Find the average number of ratings per Item
   6. List the Item_id of all items which have received rating greater than 4

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. Upload a local file called data.csv to HDFS.
   b. Verify that data.csv is uploaded.
   c. Copy data.csv to a new directory called datadir.
   d. Verify the file contents in datadir.

2. Consider a Book text file. Write a Map Reduce program to find word count. [10]

3. Solve the following using HIVE [10]
   Write HQL queries for the following
   1. Create a database with the name IRating
   2. Use the above created database
   3. Create table Rating with the fields(userid, item_id, rating, timestamp) where fields are terminated by '\t'
   4. Load data from local storage in the path home/user/data/Item_Rating"
   5. Display first 100 rows of the table
   6. List the Item_id of all items which have received rating greater than 4

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. Create two files called file1.txt and file2.txt in HDFS.
   b. Concatenate file1.txt and file2.txt into a new file called aggregated.txt.
   c. Verify the contents of aggregated.txt.
   d. Copy the file from local drive to Hadoop

2. Consider the Movie Data set. Write a Map Reduce program to find average rating per user. [10]

3. Solve the following using HIVE [10]
   Write HQL queries for the following
   1. Create a database with the name StudDetail
   2. Use the above created database.
   3. Create table Student with the fields(Stident_id, age, class, Subject_opted, marks) where fields are terminated by ','
   4. Load data from local storage in the path home/user/data/stud_marks"
   5. Find the total number of students per class.
   6. Find subject wise maximum marks

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. List the files in the root directory.
   b. Create a new directory called "test".
   c. Change into the "test" directory.
   d. Create a subdirectory called "subtest".

2. Considerthe Temperature Data set. Write a Map Reduce program to find minimum temperature for each weather station. [10]

3. Solve the following using HIVE [10]
   Write HQL queries for the following
   1. Create a database with the name StudDetail
   2. Use the above created database.
   3. Create table Student with the fields(Stident_id, age, class, Subject_opted, marks) where fields are terminated by ','
   4. Load data from local storage in the path home/user/data/stud_marks
   5. Find the average marks per subject
   6. Find subject wise maximum marks

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. List the files in the /user/data directory using an absolute path.
   b. Change into the /user/data/subdir directory using an absolute path.
   c. List the files in the current directory.
   d. Change into the root directory.

2. Considerthe Temperature Data set. Write a Map Reduce program to find maximum temperature for each weather station. [10]

3. Solve the following using Pig [10]
   Consider the employee.csv file in the format (eno, ename,salary, dept, age)
   Write PIG script for the following
   a. Load a file "employee.csv".
   b. Display all employees of ComputerDepartment.
   c. Dump all employees whose age is lessthan 30.
   d. Group the above relation by department.
   e. Dump employee having maximumage.

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. Create a new directory called "data" in HDFS.
   b. Upload a file called "sample.txt" from the local file system to the "data" directory.
   c. List the files in the "data" directory.
   d. Copy the "sample.txt" file to a new file called "sample_copy.txt" in the same directory.

2. Consider the Customer order Data set. Write a Map Reduce program to find the amount spent by each customer. [10]

3. Solve the following using Pig [10]
   Consider the Movie.csv file in the format (Movie_id,M_title, Release_year, Votes, score)
   Write PIG script for the following
   a. Load a file "Movie.csv".
   b. Dump titles of movies that havea score higher than 8.
   c. Dump the title and year of moviesmade in year 1980.
   d. Dump the top 5 highest vote-getting movies (use ORDER and LIMIT).
   e. Group Movie relation by votes

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. Create a new directory called "input" and another called "output".
   b. Upload a file called "data.csv" to the "input" directory.
   c. Copy the "data.csv" file to the "output" directory.
   d. Verify the file permissions of "data.csv".

2. Consider the Movie Data set. Write a Map Reduce program to find average rating per user. [10]

3. Solve the following using Pig [10]
   Consider the employee.csv file in the format (eno, ename,salary, dept, age)
   Write PIG script for the following
   a. Load a file "employee.csv"
   b. Group the above relation by salary.
   c. Display all employees of ComputerDepartment.
   d. Display department wise sum of salary.
   e. Dump all employees according todescending order of employee name.

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. List the files in the root directory.
   b. Create a new directory called "test".
   c. Change into the "test" directory.
   d. Create a subdirectory called "subtest".

2. Consider the Movie Data set. Write a Map Reduce program to find average rating per user. [10]

3. Solve the following using Pig [10]
   Consider the employee.csv file in the format (eno, ename,salary, dept, age)
   Write PIG script for the following
   a. Load the file "employee.csv"
   b. Display employee having maximumage.
   c. Dump department wise sum of salary.
   d. Dump all employees according to descending order of employee name.
   e. Display employees of Sales department

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. Create a new file called file1.txt in HDFS.
   b. Copy file1.txt to a new file called file2.txt.
   c. Copy file2.txt to a new directory called copydir.
   d. Verify the file contents in copydir.

2. Consider a book txt file. Write a Map Reduce program to display word count in sorted order. [10]

3. A Solve the following using Pig [10]
   Consider the Movie.csv file in the format (Movie_id,M_title, Release_year, Votes, score)
   Write PIG script for the following
   a. Load the file "Movie.csv".
   b. Dump the titles of movies that havemore than 100 votes.
   c. Dump the title and votes of movies madein year 2020.
   d. Display the votes and score of the movie"Bhoot"
   e. Group Movie relation by votes

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. Create a new directory called "input" and another called "output".
   b. Upload a file called "data.csv" to the "input" directory.
   c. Copy the "data.csv" file to the "output" directory.
   d. Rename the "data.csv" file in the "output" directory to "processed_data.csv".

2. Consider the Movie Data set. Write a Map Reduce program to find average rating per Movie. [10]

3. Solve the following using Pig [10]
   Consider the student.csv file in the format (Rollno, Sname, age, class, phone, city)

   Write PIG script for the following
   a. Load the file "student.csv"
   b. Write a pig script to group the above relation by age.
   c. Write a pig script to display all employees of ComputerDepartment.
   d. Write a pig script do display class wise number of students.
   e. Write a pig script to display all students according todescending order of student name.

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. List the files in the /user/data directory using an absolute path.
   b. Change into the /user/data/subdir directory using an absolute path.
   c. List the files in the current directory.
   d. Change into the root directory.

2. Consider a book txt file. Write a Map Reduce program to display word count in sorted order. [10]

3. Solve the following using Pig [10]
   Consider the employee.csv file in the format (eno, ename, salary, dept, age)
   Write Pig script to do the following
   a. Load the data from csv file.
   b. Dump all employees of Computer Department.
   c. Group the employees by department
   d. Dump department wise employee having maximum age.
   e. Dump department wise sum of salary.

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. Create a new directory called "input" and another called "output".
   b. Upload a file called "data.csv" to the "input" directory.
   c. Copy the "data.csv" file to the "output" directory.
   d. Verify the file permissions of "data.csv".

2. Consider a Book text file. Write a Map Reduce program to find word which has occurred maximum number of times. [10]

3. Solve the following using Pig [10]
   Write Pig script to do the following
   a. Write a pig statement to load a file "student_data.txt" containing the fields : id, firstname, lastname, age, phone, city) where fields are separated by ",".
   b. Display the above loaded relation on to the screen
   c. Describing the relation
   d. Group the above created relation by age
   e. Group the above relation by all columns
   f. Find city wise average age

4. Viva [3]

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   b. Upload a file called "sample.txt" from the local file system to the "data" directory.
   c. List the files in the "data" directory.
   d. Copy the "sample.txt" file to a new file called "sample_copy.txt" in the same directory.

2. Consider the Temperature data set. Write a Map Reduce program to find the minimum temperature at each weather station. [10]

3. A Solve the following using Pig [10]
   Write PIG script for the following
   Consider the Movie.csv file in the format
   (Movie_id,M_title, Release_year, Votes, score)
   a. Load the file "Movie.csv".
   b. Dump the titles of movies that have more than 100 votes
   c. Dump the title and votes of movies made in year 2020.
   d. Dump the votes and score of the movie"Bhoot"
   e. Group Movie relation by votes

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. List the files in the root directory.
   b. Create a new directory called "test".
   c. Change into the "test" directory.
   d. Create a subdirectory called "subtest".

2. Consider the Book text file. Write a MapReduce program to count the frequency of each word.

3. Solve the following using Pig [10]
   Write Pig script to do the following
   a. Load a file "student_data.txt" containing the fields : id, firstname, lastname, age, phone, city) where fields are separated by ",".
   b. Display the above loaded relation on to the screen
   c. Describing the relation
   d. Group the above created relation by age
   e. Group the above relation by all columns
   f. Find city wise average age

4. Viva [3]

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1. Write Hadoop shell commands to do the following [2]
   a. Create a new directory called "input" and another called "output".
   b. Upload a file called "data.csv" to the "input" directory.
   c. Copy the "data.csv" file to the "output" directory.
   d. Rename the "data.csv" file in the "output" directory to "processed_data.csv".

2. Consider the Friends data set. Write a MapReduce program to find the person having maximun number of friends. [10]

3. Solve the following using Pig [10]
   Write PIG script for the following
   Consider the employee.csv file in the format (eno, ename, salary, dept, age)
   a. Load a file "employee.csv".
   b. Display all employees of Computer Department.
   c. Dump all employees whose age is less than 30.
   d. Group the above relation by department.
   e. Dump employee having maximum age for each department.

4. Viva [3]